Questions: Introduction to analysis of variance (ANOVA)

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Summary

A selection of questions to test your understanding of the procedure of Analaysis of Variance.

Before attempting these questions it is highly recommended that you read Guide: Introduction to ANOVA.

Q1

This question is about the logic behind one-way ANOVA.

1.1

What does (one-way) ANOVA check for?

1.2

State the key assumptions of (one-way) ANOVA.

1.3

Which assumption is critical to the test and why?

1.4

If the other assumptions are not fulfilled what alternatives can you use (and what do these alternatives require to be valid as well)?

Q2

This question is about the mathematical side of ANOVA.

2.1.

You define the F-test statistic as:

$$f_0 = \frac{s_G^2}{s_W^2}$$

explain the meaning of s_G^2 and s_W^2. You can also provide the formulae for calculating these parameters to expand your interpretation.

2.2

Now focus on the F-test statistic itself, what does it represent? If it is noticeably large might it indicate the outcome of the hypothesis test? Conversely, if it small - what does it tell you then? In each case state if you reject or fail to reject your null hypothesis (H_0)?

2.3

You want to conduct an ANOVA analysis for n data points and k groups. Write down the F distribution against which you would compare the f test statistic (assume that you test at 95% confidence level).

2.4

Now you learn that there are in fact 8 groups and 75 data points across all of them, which leads to your reference F distribution quantile being 2.482985. In addition, you know that $s_G^2 = 8.78901$, and $s_W^2 = 1.3729$. Calculate the F test statistic and provide an interpretation of its outcome.

Q3

This takes you through the whole process of conducting the ANOVA analysis. It also briefly touches upon post-ANOVA tests so you should read the Facthsheet: Family-wise error

Cantor's Confectionery wishes to compare the mean sales-to-production ratio across eight flavours.

If the mean is negative, then the amount of the ice cream sold on that day was smaller than the amount of ice cream produced. If the mean is positive, then the amount of ice cream sold on that day was larger than the amount of ice cream produced.

They obtain the following data across 15 randomly selected days during the summer:

Sample size	Mean ($\bar{x}$)	Standard deviation (σ)
15	-0.550	1.272
15	-0.497	0.760
15	-0.053	0.960
15	-0.012	1.016
15	-0.228	0.964
15	0.576	1.309
15	0.474	1.020
15	0.011	1.280

3.1

State why you would prefer to use ANOVA, rather than other hypothesis tests (like Z or t test).

3.2

State the null (H_0) and the alternative (H_1) hypothesis under which you conduct the test on this data.

3.3

Input the data into the ANOVA calculator and conduct the ANOVA analysis.

```
#| '!! shinylive warning !!': |
#|   shinylive does not work in self-contained HTML documents.
#|   Please set `embed-resources: false` in your metadata.
#| standalone: true
#| viewerHeight: 800
library(shiny)
library(bslib)

ui <- page_sidebar(
  title = "ANOVA Calculator",
  sidebar = sidebar(
    numericInput("n_groups", "Number of Groups", value = 3, min = 2, max = 10),
    actionButton("update_inputs", "Update Input Fields", class = "btn-primary mt-2"),
```

```
    hr(),
    uiOutput("group_inputs"),
    actionButton("calculate", "Calculate ANOVA", class = "btn-success mt-3")
  ),
  card(
    card_header("ANOVA Results"),
    verbatimTextOutput("anova_results")
  ),
  card(
    card_header("Summary Statistics"),
    tableOutput("summary_table")
  ),
  layout_columns(
    col_widths = c(6, 6),
    card(
      card_header("Group Means Plot"),
      plotOutput("means_plot")
    ),
    card(
      card_header("Distribution Plot"),
      plotOutput("distribution_plot")
    )
  ),
  card(
    card_header("Suggested Interpretation"),
    uiOutput("interpretation")
  )
)

server <- function(input, output, session) {

  # Generate dynamic inputs for each group
  output$group_inputs <- renderUI({
    req(input$update_inputs)
    n_groups <- isolate(input$n_groups)

    lapply(1:n_groups, function(i) {
      tagList(
```

```
      h5(paste("Group", i)),
      numericInput(paste0("mean_", i), "Mean:", value = 50, step = 0.1),
      numericInput(paste0("n_", i), "Sample Size:", value = 10, min = 2),
      numericInput(paste0("sd_", i), "Standard Deviation:", value = 5, min = 0.01, s
      hr()
    )
  })
})

# Calculate ANOVA when button is clicked
anova_data <- eventReactive(input$calculate, {
  n_groups <- input$n_groups

  # Collect data from inputs
  means <- numeric(n_groups)
  sample_sizes <- numeric(n_groups)
  sds <- numeric(n_groups)

  for (i in 1:n_groups) {
    means[i] <- input[[paste0("mean_", i)]]
    sample_sizes[i] <- input[[paste0("n_", i)]]
    sds[i] <- input[[paste0("sd_", i)]]
  }

  # Validate inputs
  if (any(is.na(means)) || any(is.na(sample_sizes)) || any(is.na(sds))) {
    return(NULL)
  }

  # Calculate ANOVA from summary statistics
  N <- sum(sample_sizes)
  k <- n_groups
  grand_mean <- sum(means * sample_sizes) / N
  SS_between <- sum(sample_sizes * (means - grand_mean)^2)
  SS_within <- sum((sample_sizes - 1) * sds^2)
  SS_total <- SS_between + SS_within
  df_between <- k - 1
  df_within <- N - k
```

```
  df_total <- N - 1
  MS_between <- SS_between / df_between
  MS_within <- SS_within / df_within
  F_stat <- MS_between / MS_within
  p_value <- pf(F_stat, df_between, df_within, lower.tail = FALSE)

  list(
    means = means,
    sample_sizes = sample_sizes,
    sds = sds,
    N = N,
    k = k,
    grand_mean = grand_mean,
    SS_between = SS_between,
    SS_within = SS_within,
    SS_total = SS_total,
    df_between = df_between,
    df_within = df_within,
    df_total = df_total,
    MS_between = MS_between,
    MS_within = MS_within,
    F_stat = F_stat,
    p_value = p_value
  )
})

# Display ANOVA table
output$anova_results <- renderPrint({
  data <- anova_data()
  req(data)

  cat("ANOVA Table\n")
  cat("===========================================================\n")
  cat(sprintf("%-20s %10s %10s %12s %12s\n", "Source", "df", "SS", "MS", "F"))
  cat("-----------------------------------------------------------\n")
  cat(sprintf("%-20s %10d %10.4f %12.4f %12.4f\n",
              "Between Groups", data$df_between, data$SS_between,
              data$MS_between, data$F_stat))
```

```
  cat(sprintf("%-20s %10d %10.4f %12.4f\n",
              "Within Groups", data$df_within, data$SS_within, data$MS_within))
  cat(sprintf("%-20s %10d %10.4f\n",
              "Total", data$df_total, data$SS_total))
  cat("============================================================\n\n")
  cat(sprintf("F-statistic: %.4f\n", data$F_stat))
  cat(sprintf("p-value: %.6f\n", data$p_value))
  cat(sprintf("Significance level: %s\n",
              ifelse(data$p_value < 0.001, "p < 0.001 ***",
                     ifelse(data$p_value < 0.01, "p < 0.01 **",
                            ifelse(data$p_value < 0.05, "p < 0.05 *",
                                   "Not significant")))))
})

# Display summary statistics table
output$summary_table <- renderTable({
  data <- anova_data()
  req(data)

  data.frame(
    Group = paste("Group", 1:data$k),
    Mean = round(data$means, 3),
    SD = round(data$sds, 3),
    N = data$sample_sizes
  )
}, striped = TRUE, hover = TRUE)

# Plot group means with error bars (base R, no color coding)
output$means_plot <- renderPlot({
  data <- anova_data()
  req(data)

  se <- data$sds / sqrt(data$sample_sizes)

  par(mar = c(5, 5, 4, 2))
  bp <- barplot(data$means, names.arg = paste("Group", 1:data$k),
                col = "lightgray", ylim = c(0, max(data$means + se) * 1.2),
                ylab = "Mean Value", xlab = "Group",
```

```
                main = "Group Means with Standard Error Bars",
                cex.main = 1.2, cex.lab = 1.1, border = "black")

  arrows(bp, data$means - se, bp, data$means + se,
         angle = 90, code = 3, length = 0.1, lwd = 2)

  abline(h = data$grand_mean, col = "black", lty = 2, lwd = 2)
  text(max(bp), data$grand_mean,
       labels = paste("Grand Mean =", round(data$grand_mean, 2)),
       pos = 3, col = "black", cex = 1.1)
})

# Plot distributions (base R, no color coding)
output$distribution_plot <- renderPlot({
  data <- anova_data()
  req(data)

  # Generate simulated data
  all_data <- list()
  for (i in 1:data$k) {
    all_data[[i]] <- rnorm(data$sample_sizes[i], data$means[i], data$sds[i])
  }

  par(mar = c(5, 5, 4, 2))
  boxplot(all_data, names = paste("Group", 1:data$k),
          col = "lightgray", ylab = "Value", xlab = "Group",
          main = "Distribution of Values by Group",
          cex.main = 1.2, cex.lab = 1.1, pch = 16, border = "black")

  # Add individual points
  for (i in 1:data$k) {
    points(jitter(rep(i, length(all_data[[i]])), amount = 0.1),
           all_data[[i]], pch = 16, col = rgb(0, 0, 0, 0.3), cex = 0.8)
  }

  mtext("Note: Data points are simulated from the summary statistics",
        side = 1, line = 4, cex = 0.9, font = 3)
})
```

```
  # Display interpretation
  output$interpretation <- renderUI({
    data <- anova_data()
    req(data)

    if (data$p_value < 0.05) {
      tagList(
        p(strong("Conclusion:"), "There is a statistically significant difference bet
          (F(", data$df_between, ", ", data$df_within, ") = ", round(data$F_stat, 3),
          ", p = ", format.pval(data$p_value, digits = 3), ")."),
        p("This suggests that at least one group mean is significantly different from
      )
    } else {
      tagList(
        p(strong("Conclusion:"), "There is no statistically significant difference be
          (F(", data$df_between, ", ", data$df_within, ") = ", round(data$F_stat, 3),
          ", p = ", format.pval(data$p_value, digits = 3), ")."),
        p("The data does not provide sufficient evidence to conclude that the group me
      )
    }
  })
}

shinyApp(ui, server)
```

3.4

Report the F-test statistic (f_0) and the associated p-value. Under 99% confidence level what is the outcome of the tests you ran? (Hint: $\alpha = 0.01$ now).

3.5

If you increase the confidence level to 95% what is the outcome of the test now - has it changed? What does it tell you about the conclusions drawn earlier?

3.6

Finally, you have to present the conclusions of your statistical analysis to the owner of the Confectionery - Mr. Cantor. Write a brief report, not using statistical terminology, to summarize your findings and what have you learned about the sales-to-production ratio across eight of his ice cream flavours?

Q4

This question concerns completing the ANOVA analysis.

You obtain the following data of the customer-to-cashier ratio per hour at Cantor's confectionery across four randomly chosen hours on four randomly chosen days in a month:

Day	Customer-to-cashier ratio
1	$13, 12, 15, 8, 10$
2	$11, 13, 12, 14, 9$
3	$15, 19, 20, 13, 2$
4	$14, 15, 12, 17, 13$

You also obtain the mean and standard deviation of the customer-to-cashier ratio for each day:

Day	Mean ($\bar{x}$)	Standard deviation
1	11.6	2.701851
2	11.8	1.923538
3	13.8	7.190271
4	14.2	1.923538

4.1

Conduct a one-way ANOVA analysis using the ANOVA calculator.

```
#| '!! shinylive warning !!': |
#|   shinylive does not work in self-contained HTML documents.
#|   Please set `embed-resources: false` in your metadata.
#| standalone: true
#| viewerHeight: 800
library(shiny)
```

```r
library(bslib)

ui <- page_sidebar(
  title = "ANOVA Calculator",
  sidebar = sidebar(
    numericInput("n_groups", "Number of Groups", value = 3, min = 2, max = 10),
    actionButton("update_inputs", "Update Input Fields", class = "btn-primary mt-2"),
    hr(),
    uiOutput("group_inputs"),
    actionButton("calculate", "Calculate ANOVA", class = "btn-success mt-3")
  ),
  card(
    card_header("ANOVA Results"),
    verbatimTextOutput("anova_results")
  ),
  card(
    card_header("Summary Statistics"),
    tableOutput("summary_table")
  ),
  layout_columns(
    col_widths = c(6, 6),
    card(
      card_header("Group Means Plot"),
      plotOutput("means_plot")
    ),
    card(
      card_header("Distribution Plot"),
      plotOutput("distribution_plot")
    )
  ),
  card(
    card_header("Suggested Interpretation"),
    uiOutput("interpretation")
  )
)

server <- function(input, output, session) {
```

```
# Generate dynamic inputs for each group
output$group_inputs <- renderUI({
  req(input$update_inputs)
  n_groups <- isolate(input$n_groups)

  lapply(1:n_groups, function(i) {
    tagList(
      h5(paste("Group", i)),
      numericInput(paste0("mean_", i), "Mean:", value = 50, step = 0.1),
      numericInput(paste0("n_", i), "Sample Size:", value = 10, min = 2),
      numericInput(paste0("sd_", i), "Standard Deviation:", value = 5, min = 0.01,
      hr()
    )
  })
})

# Calculate ANOVA when button is clicked
anova_data <- eventReactive(input$calculate, {
  n_groups <- input$n_groups

  # Collect data from inputs
  means <- numeric(n_groups)
  sample_sizes <- numeric(n_groups)
  sds <- numeric(n_groups)

  for (i in 1:n_groups) {
    means[i] <- input[[paste0("mean_", i)]]
    sample_sizes[i] <- input[[paste0("n_", i)]]
    sds[i] <- input[[paste0("sd_", i)]]
  }

  # Validate inputs
  if (any(is.na(means)) || any(is.na(sample_sizes)) || any(is.na(sds))) {
    return(NULL)
  }

  # Calculate ANOVA from summary statistics
  N <- sum(sample_sizes)
```

```
  k <- n_groups
  grand_mean <- sum(means * sample_sizes) / N
  SS_between <- sum(sample_sizes * (means - grand_mean)^2)
  SS_within <- sum((sample_sizes - 1) * sds^2)
  SS_total <- SS_between + SS_within
  df_between <- k - 1
  df_within <- N - k
  df_total <- N - 1
  MS_between <- SS_between / df_between
  MS_within <- SS_within / df_within
  F_stat <- MS_between / MS_within
  p_value <- pf(F_stat, df_between, df_within, lower.tail = FALSE)

  list(
    means = means,
    sample_sizes = sample_sizes,
    sds = sds,
    N = N,
    k = k,
    grand_mean = grand_mean,
    SS_between = SS_between,
    SS_within = SS_within,
    SS_total = SS_total,
    df_between = df_between,
    df_within = df_within,
    df_total = df_total,
    MS_between = MS_between,
    MS_within = MS_within,
    F_stat = F_stat,
    p_value = p_value
  )
})

# Display ANOVA table
output$anova_results <- renderPrint({
  data <- anova_data()
  req(data)
```

```
  cat("ANOVA Table\n")
  cat("============================================================\n")
  cat(sprintf("%-20s %10s %10s %12s %12s\n", "Source", "df", "SS", "MS", "F"))
  cat("------------------------------------------------------------\n")
  cat(sprintf("%-20s %10d %10.4f %12.4f %12.4f\n",
              "Between Groups", data$df_between, data$SS_between,
              data$MS_between, data$F_stat))
  cat(sprintf("%-20s %10d %10.4f %12.4f\n",
              "Within Groups", data$df_within, data$SS_within, data$MS_within))
  cat(sprintf("%-20s %10d %10.4f\n",
              "Total", data$df_total, data$SS_total))
  cat("============================================================\n\n")
  cat(sprintf("F-statistic: %.4f\n", data$F_stat))
  cat(sprintf("p-value: %.6f\n", data$p_value))
  cat(sprintf("Significance level: %s\n",
              ifelse(data$p_value < 0.001, "p < 0.001 ***",
                     ifelse(data$p_value < 0.01, "p < 0.01 **",
                            ifelse(data$p_value < 0.05, "p < 0.05 *",
                                   "Not significant")))))
})

# Display summary statistics table
output$summary_table <- renderTable({
  data <- anova_data()
  req(data)

  data.frame(
    Group = paste("Group", 1:data$k),
    Mean = round(data$means, 3),
    SD = round(data$sds, 3),
    N = data$sample_sizes
  )
}, striped = TRUE, hover = TRUE)

# Plot group means with error bars (base R, no color coding)
output$means_plot <- renderPlot({
  data <- anova_data()
  req(data)
```

```
  se <- data$sds / sqrt(data$sample_sizes)

  par(mar = c(5, 5, 4, 2))
  bp <- barplot(data$means, names.arg = paste("Group", 1:data$k),
                col = "lightgray", ylim = c(0, max(data$means + se) * 1.2),
                ylab = "Mean Value", xlab = "Group",
                main = "Group Means with Standard Error Bars",
                cex.main = 1.2, cex.lab = 1.1, border = "black")

  arrows(bp, data$means - se, bp, data$means + se,
         angle = 90, code = 3, length = 0.1, lwd = 2)

  abline(h = data$grand_mean, col = "black", lty = 2, lwd = 2)
  text(max(bp), data$grand_mean,
       labels = paste("Grand Mean =", round(data$grand_mean, 2)),
       pos = 3, col = "black", cex = 1.1)
})

# Plot distributions (base R, no color coding)
output$distribution_plot <- renderPlot({
  data <- anova_data()
  req(data)

  # Generate simulated data
  all_data <- list()
  for (i in 1:data$k) {
    all_data[[i]] <- rnorm(data$sample_sizes[i], data$means[i], data$sds[i])
  }

  par(mar = c(5, 5, 4, 2))
  boxplot(all_data, names = paste("Group", 1:data$k),
          col = "lightgray", ylab = "Value", xlab = "Group",
          main = "Distribution of Values by Group",
          cex.main = 1.2, cex.lab = 1.1, pch = 16, border = "black")

  # Add individual points
  for (i in 1:data$k) {
    points(jitter(rep(i, length(all_data[[i]])), amount = 0.1),
```

```
            all_data[[i]], pch = 16, col = rgb(0, 0, 0, 0.3), cex = 0.8)
    }

    mtext("Note: Data points are simulated from the summary statistics",
          side = 1, line = 4, cex = 0.9, font = 3)
  })

  # Display interpretation
  output$interpretation <- renderUI({
    data <- anova_data()
    req(data)

    if (data$p_value < 0.05) {
      tagList(
        p(strong("Conclusion:"), "There is a statistically significant difference bet
          (F(", data$df_between, ", ", data$df_within, ") = ", round(data$F_stat, 3),
          ", p = ", format.pval(data$p_value, digits = 3), ")."),
        p("This suggests that at least one group mean is significantly different from
      )
    } else {
      tagList(
        p(strong("Conclusion:"), "There is no statistically significant difference be
          (F(", data$df_between, ", ", data$df_within, ") = ", round(data$F_stat, 3),
          ", p = ", format.pval(data$p_value, digits = 3), ")."),
        p("The data does not provide sufficient evidence to conclude that the group m
      )
    }
  })
}

shinyApp(ui, server)
```

4.2

Report your findings: state the F-test statistic (f_0) and its associated p-value. At 99% confidence level $(\alpha = 0.01)$ interpret the results of the test.

4.3

Now look closely at the data once again, take a look at the standard deviations as well. Does anything stand out? Does anything in the data make you question the underlying assumptions for the test?

After attempting the questions above, please click this link to find the answers.

Version history and licensing

v1.0: initial version created 04/26 by Jakub Brzozowski as a part of a University of St Andrews VIP project.